# 1. Twist symmetry in quartic extension index calculus

22 October 2026. Full source-linked record: `QUARTIC_TWIST24_REPORT.md`.

## 1.1. Question and protocol

We tested a transported Frobenius on a quadratic twist over $F_{p^4}$, combined it with the cubic $j = 0$ automorphism, and measured an unchanged physical base, the union of two invariant lines, the full invariant two-dimensional abscissa space, tagged pair indexes, and dense same-residual rank extraction. The line panel has seven positive toy curves and three seeds; the full 2D target solves use $p = 13, 37$. Each run has one public target $[271]G$. The signed four-factor oracle, matrix encodings, and 24-fold rho reference are native Rust. The frozen protocol and raw records are in the same study directory.

The construction uses $F = \frac{F_p[t]}{t^4-d}$ with $d$ the least nonsquare, $E : y^2 = x^3 + b_0 t^3$, $h = t^{\frac{1-p}{2}}$, $\omega^3 = 1$, and $\psi(x, y) = (h^2 x^p, h^3 y^p)$, $\iota(x, y) = (\omega x, y)$. The norm identity $h^{1+p+p^2+p^3} = -1$ gives $\psi^4 = [-1]$. The two maps commute. On the certified prime-order group their checked eigenvalues have orders eight and three, yielding order 24 on signed points and 12 on nonzero abscissae. Thus a stable base with $B$ abscissae has $\frac{B}{12}$ unknown logs, compared with $\frac{B}{3}$ for the cubic automorphism alone.

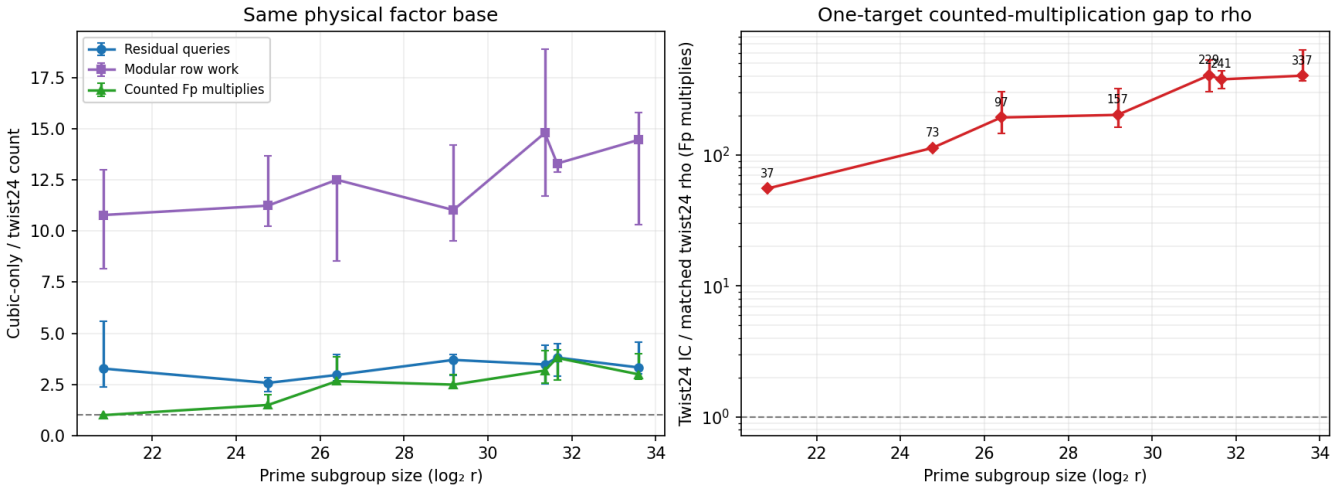
For the four lines $t^j F_p$, $\psi$ multiplies abscissae by $z^{j-1}$, $z = t^{p-1}$. Lines 1 and 3 cannot contain a nonzero odd-order point: $\psi$ or $\psi^2$ would fix its abscissa and hence act as either plus one or minus one, contradicting $\psi^4 = [-1]$. The complete sweep found only lines 0 and 2 usable, with line 0/2 counts shown below.

| $p$ | $r$ | line 0 | line 2 | 24-fold columns 0→0+2 |
|---|---|---|---|---|
| 13 | 28,753 | 0 | 0 | none |
| 37 | 1,873,633 | 36 | 12 | 3→4 |
| 73 | 28,408,033 | 48 | 48 | 4→8 |
| 97 | 88,538,353 | 36 | 36 | 3→6 |
| 157 | 607,561,153 | 108 | 84 | 9→16 |
| 229 | 2,750,120,881 | 96 | 96 | 8→16 |
| 241 | 3,373,515,841 | 96 | 96 | 8→16 |
| 337 | 12,897,718,513 | 180 | 180 | 15→30 |

## 1.2. Fixed physical base: what the fold changes

The same signed points and residual/decomposition stream feed flat, cubic-only and combined matrices. All 63 IC solves and 1,008 rho walks recovered $271$. Independent native replay reconstructs 1,821 selected relations, all row encodings and matrix spans. The full A/A rerun matched every non-timing field.

| $p$ | cubic/twist queries | cubic/twist row ops | cubic/twist $M$ | twist/rho24 $M$ |
|---|---|---|---|---|
| 37 | 3.27 | 10.78 | 0.998 | 55.5 |
| 73 | 2.57 | 11.24 | 1.49 | 112.9 |
| 97 | 2.96 | 12.51 | 2.66 | 192.9 |
| 157 | 3.70 | 11.03 | 2.49 | 201.9 |
| 229 | 3.48 | 14.79 | 3.18 | 404.6 |
| 241 | 3.81 | 13.30 | 3.80 | 376.8 |
| 337 | 3.33 | 14.45 | 2.99 | 402.1 |



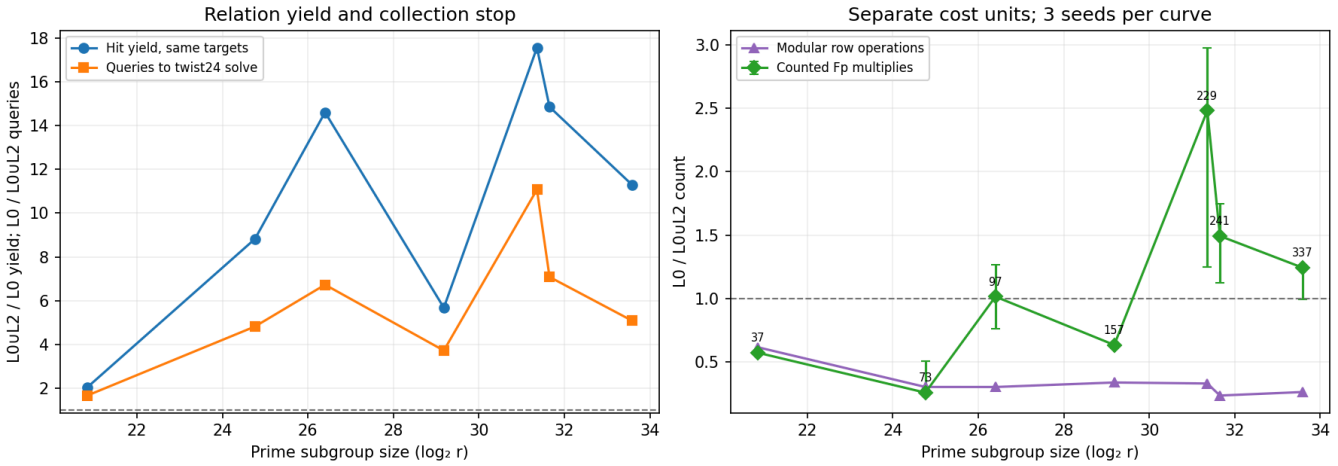

$M$ is recorded base-field multiplications on each path with its relevant model, input, eigen, base, map, table, stream and verification work. Row operations are a separate unit. The 64-query batch floor erases the field-multiplication improvement at $p = 37$. The rho comparison is same-target and includes its 24-fold endomorphism; the large gap is a partial-cost diagnostic, not an admitted speed ratio.

Applying an endomorphism to a found relation transforms the target and multiplies both the folded row and its right side by the eigenvalue. The audit verifies this covariance on every selected relation. Orbit images are not independent rows. Therefore the unchanged physical base gains relation **utility**, not raw hit probability.

### 1.3. Two invariant lines: actual relation yield

For each curve and seed, $L_0$ and $L_0 \cup L_2$ receive the exact same target stream, with identical $L_0$ points and signs. The independent native oracle recomputed every one of 69,045 query hit counts using a fully signed two-pair hash, re-added 5,298 selected group relations, and checked all 126 matrix solves. The producer recorded 7,275 decompositions. Both full runs matched on every non-timing field. Medians are over three paired, separate one-target runs.

| $p$ | union/single hit yield | single/union queries | single/union row ops | single/union counted $M$ |
|---|---|---|---|---|
| 37 | 2.04 | 1.67 | 0.61 | 0.57 |
| 73 | 8.82 | 4.82 | 0.30 | 0.26 |
| 97 | 14.60 | 6.73 | 0.30 | 1.01 |
| 157 | 5.68 | 3.72 | 0.34 | 0.63 |
| 229 | 17.55 | 11.07 | 0.33 | 2.48 |
| 241 | 14.85 | 7.08 | 0.23 | 1.49 |
| 337 | 11.28 | 5.08 | 0.26 | 1.24 |




The union improves same-target hit yield on every curve and lowers query count, but doubles or nearly doubles matrix columns and increases row work. Counted field multiplications improve in all three seeds at $p = 229, 241$; they worsen at $p = 37, 73, 157$ and straddle one at $p = 97, 337$. The figure keeps the row-operation and field-multiplication units distinct. This is a relation-yield result with a cost tradeoff, not a uniform whole-method win.

## 1.4. Interpretation, limits, and provenance

The combination of Frobenius-invariant factor bases and matrix-orbit savings is prior work (Galbraith–Granger–Merz–Petit). GLS twist endomorphisms are prior work (Galbraith–Lin–Scott). This study makes a specific canonical quartic construction and paired mixed-line measurement explicit. An earlier repository E18 run already used the cubic $j = 0$ fold; this panel provides the same-base cubic control. The earlier repository E2b GLS experiment implemented an $S_3$ Weil-descent resultant oracle. The main quartic relation panel uses exact four-sum meet-in-the-middle; a separate six-query pilot runs the native $S_5$ solver.

The experiment is L0 on a shared Intel Xeon 6975P-C host. The main panel does not emit IC-stream group additions, hash/table operations or a calibrated group-addition-equivalent band. Its field-multiplication path projection excludes work done solely for the other two matrix arms. Whole-process wall times were not isolated and are not used for a speed claim. A single-arm sealed run and full polynomial relation stream compared with msolve are needed before an end-to-end cost or asymptotic claim. The four one-dimensional Frobenius eigenspaces were exhausted. A separate full census tested the two-dimensional space $F_p + t^2 F_p$; other invariant subspaces and curve models remain open.

The field/twist proof above is handwritten and computationally checked. `QuarticTwistSymmetry.lean` proves conditional consequences (fourth-to-eighth iterate, composed eigenvalue, row covariance) with an audited Lean kernel. It does not formalize the concrete finite field or twist isomorphism. The full proof map and source links are in `QUARTIC_TWIST24_REPORT.md`.

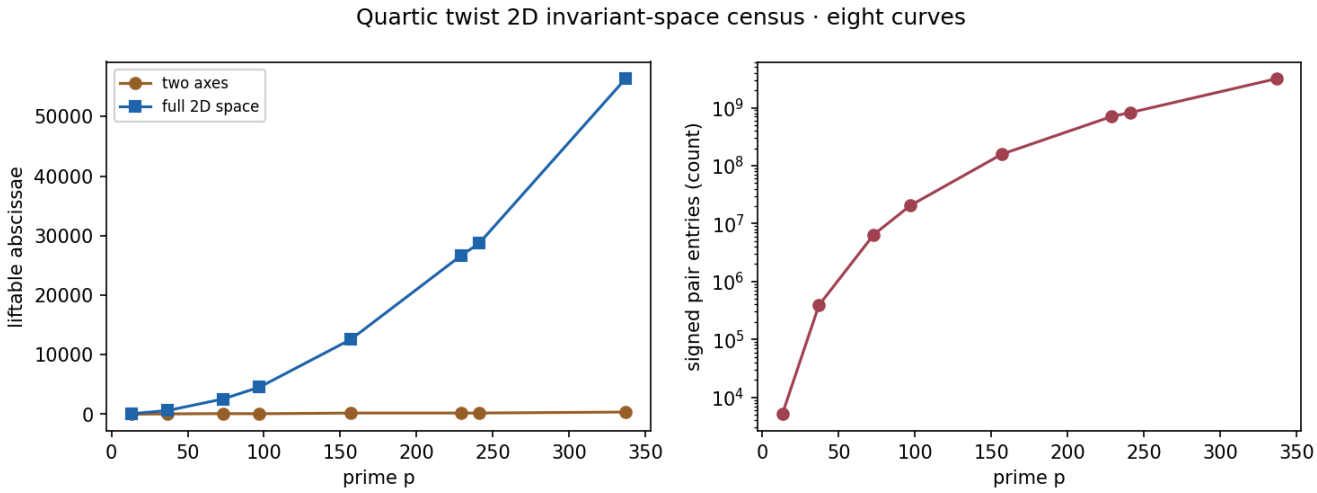
Evidence: `quartic_twist24_full_v2.ndjson`, `quartic_twist24_native_audit_v2.json`, `quartic_twist_mixed_v1.ndjson`, `quartic_twist_mixed_native_audit_v1.json`, `quartic_twist_mixed_summary_v1.json`, the frozen protocol, and `OPEN_PROBLEMS.md`. Write-once run reports with graphs and PDFs are under `reports/`. All artifacts are in this study directory.

## 1.5. Polynomial pilot and larger invariant space

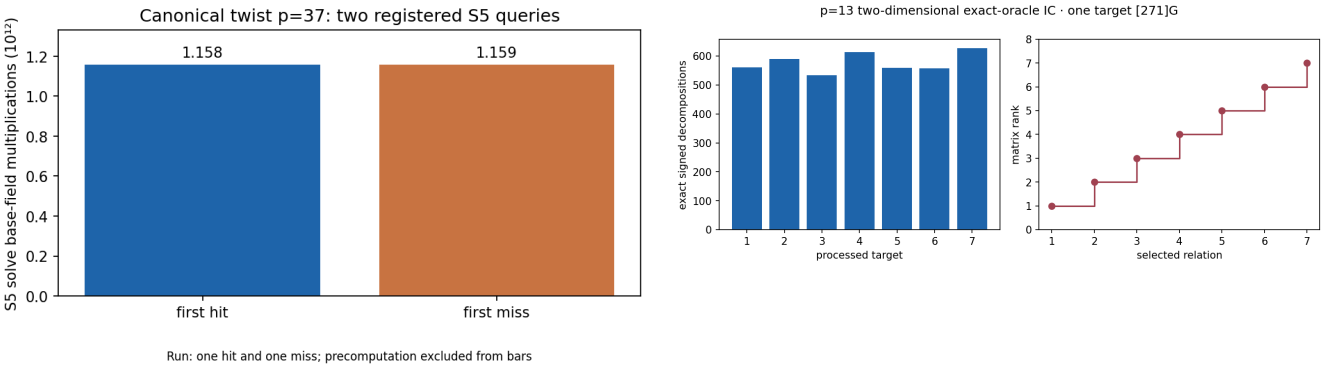
The registered native $S_5$ pilot ran one successful and one unsuccessful line-0 residual at each $p = 37, 73, 97$. All six polynomial outputs exactly matched the signed group oracle; every returned quadruple passed group addition. Each curve's symmetrisation precomputation used 1.226 billion $F_p$ multiplications. The same degree-29 Macaulay matrix shape appeared on all six solves: 41,780 rows and 40,920 columns. The plot shows counted solve multiplications; shared-host L0 solve times were 49–58 seconds and are not isolated speed evidence.

| $p$ | hit query / $10^{12}M$ | miss query / $10^{12}M$ | oracle agreement |
|---|---|---|---|
| 37 | 0 / 1.158 | 1 / 1.159 | 2/2 |
| 73 | 25 / 1.203 | 0 / 1.189 | 2/2 |
| 97 | 14 / 1.211 | 0 / 1.197 | 2/2 |

The two-dimensional invariant abscissa space $F_p + t^2 F_p$ was exhaustively scanned on all eight curves. Independent norm-character and point-orbit replay checked all 265,008 candidate abscissae. Every nonempty abscissa orbit has size 12. The formerly empty-line $p = 13$ curve has 72 liftable interior abscissae and six folded columns. At $p = 337$, the direct exact-oracle pair construction would generate 3.178 billion signed entries; this count does not measure allocated memory.




On the $p = 13$ two-dimensional base, a separately frozen one-target exact-oracle IC pilot processed seven queries, found 4,041 decompositions, selected seven verified relations, reached rank seven, and recovered $271$. Independent replay recomputed all query counts and rows; the full A/A rerun matched every non-timing field. This shows that the larger invariant base revives the toy empty-line case. It is not a polynomial-system or rho speed result.



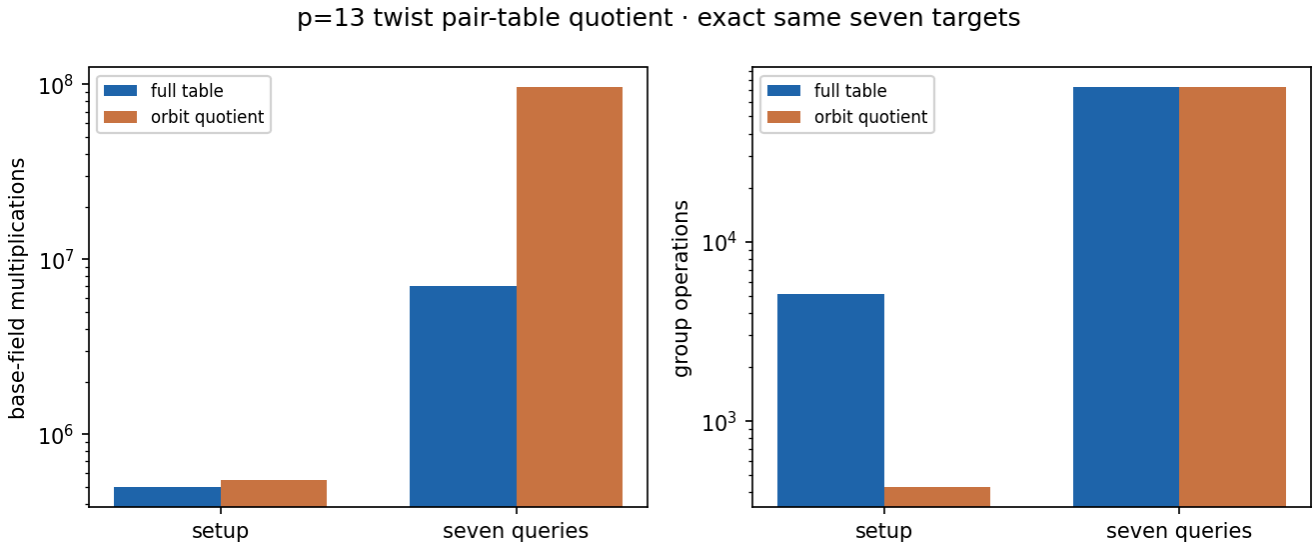

The complete measurements, per-curve S5 records, two-dimensional census, native audits, p=13 run, figures and write-once run reports are linked from `QUARTIC_TWIST24_REPORT.md`.

The later registered mixed-component polynomial experiment is summarized in the final section of this PDF and in `MIXED_POLYNOMIAL_REPORT.pdf`; the six-query S5 pilot above remains its historical single-line baseline.

### 1.6. Twist symmetry in the relation-finder pair table

On the same p=13 two-dimensional base, the signed two-factor table has 5,112 entries modulo global sign. The 12 abscissa actions partition it into 426 orbits, each of size 12. The direct quotient stores one pair representative per orbit under 350 canonical sum-abscissa keys. It reconstructs transformed pairs when queried. On every one of the seven previously recorded residuals, its **complete signed four-factor decomposition set** equals that of the full pair table; all 4,041 decompositions and seven selected IC relations remain available. A full A/A rerun matched every non-timing field.

| table arm | stored pair records | setup $F_pM$ | seven-query $F_pM$ | setup group ops |
|---|---|---|---|---|
| full | 5,112 | 495,864 | 7,029,782 | 5,112 |
| orbit quotient | 426 | 544,746 | 96,036,422 | 426 |



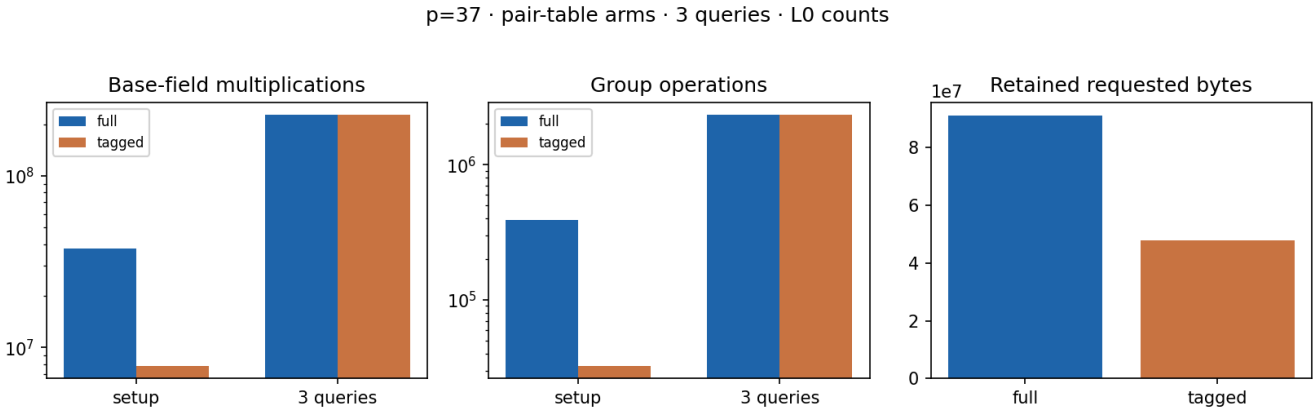

This exact symmetry quotient reduces stored pair **records** by 12 times and setup group additions by 12 times. It does not measure allocated bytes; action maps and hash buckets also consume memory. Its canonical lookup uses 13.66 times the query base-field multiplications of the full table, while query group-operation counts are equal. Therefore the tested direct quotient preserves relation yield and trades arithmetic for a smaller representative set. It is not a relation-lookup speed improvement on these targets.

Raw records: `quartic_twist_pair_quotient_p13_v1.json` and its A/A copy. Source: `examples/quartic_twist_pair_quotient_p13.rs`; the frozen protocol, full run report and exact operation counts are linked in `QUARTIC_TWIST24_REPORT.md`.

### 1.7. Action-tagged relation-finder index

The ordered, hash-key and direct-posting variants store one pair tuple per 12-action orbit while indexing each actual transformed pair-sum abscissa. They avoid query-time canonicalization. On the same p=13 seven residuals and p=37 three residuals, every variant returned the **exact same complete signed four-factor decomposition sets** as the full table. A separate group addition audited all 5,112 and 388,752 postings. Six non-timing A/A diffs are empty.

| $p$ | full / tagged setup $F_pM$ | full / tagged setup group ops | full / tagged query $F_pM$ | full / direct retained requested MB |
|---|---|---|---|---|
| 13 | 495,864 / 154,922 | 5,112 / 426 | 7,029,782 / 7,029,782 | 1.273 / 0.682 |
| 37 | 37,708,944 / 7,796,605 | 388,752 / 32,396 | 226,616,754 / 226,616,754 | 91.007 / 47.721 |



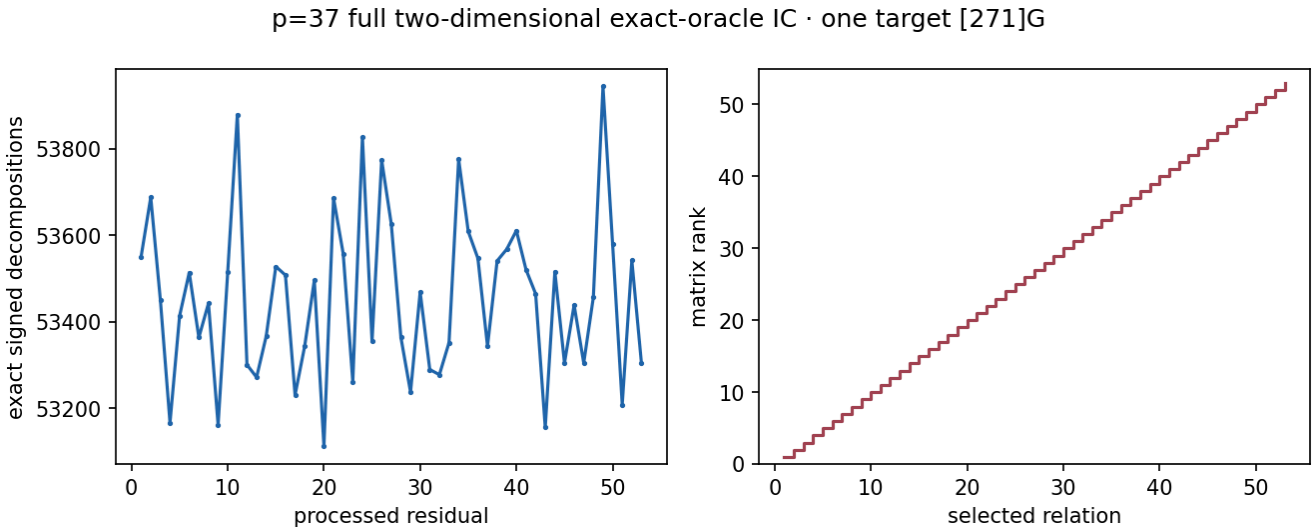

Setup group additions fall 12-fold; counted setup $F_p$ multiplications fall 3.20-fold at p=13 and 4.84-fold at p=37. Query $F_p$ multiplication and group-operation counts are exactly equal. For setup plus these registered queries, direct posting uses 1.047-fold and 1.128-fold fewer counted $F_p$ multiplications. The ordered index has the smallest retained requested payload but high binary-search overhead; the hash and direct variants use more bytes and less query CPU. Shared-host L0 timings do not establish a wall-time gain. Retained requested bytes are allocator payloads, not RSS. A complete one-target polynomial index calculus, rho comparison and isolated native-time result remain open.

The twist action alone does not create new relations on a fixed physical base. Exact decomposition sets are identical pointwise; physical-base expansion, as in the mixed-line and 2D experiments, is what raises hit yield. The tagged index changes the **cost** of finding the same relations. Commands, source digests, all raw counters and write-once run PDFs are linked from `QUARTIC_TWIST24_REPORT.md`.

### 1.8. Full p=37 two-dimensional rank pilot

The registered full $F_p + t^2 F_p$ base has 624 signed physical points and 52 twist-folded factor-log columns. The first 53 seed-zero residuals on one public target $[271]G$ yielded 2,833,126 exact signed decompositions and 53 selected relations. The matrix reached rank 53 and recovered 271; every factor log was checked by group multiplication. A separate point-keyed four-sign pair oracle enumerated all 1,873,633 subgroup logs, recomputed every hit count, checked selected rows and rank, and recovered the same scalar. The non-timing A/A diff is empty.

| phase | $F_p$ multiplications | group operations |
|---|---|---|
| model and base | 2,568,538 | 0 |
| input, eigenvalues, orbit map | 1,898,403 | 17,726 |
| full pair table | 37,708,944 | 388,752 |
| 53-query stream | 4,003,923,826 | 41,277,322 |
| answer checks | 2,004,307 | 18,854 |

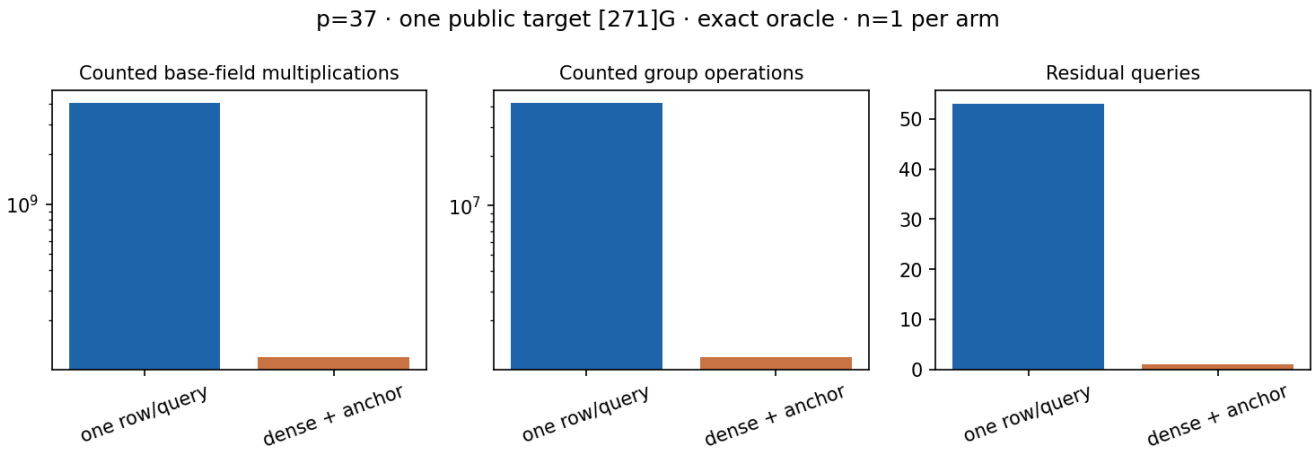



The two-line shared prefix had 43 hits and 89 decompositions over these 53 residuals; the full 2D base had 53 hits and 2,833,126 decompositions. The larger base also raises the factor-log column count and per-query pair work. Total counted path: 4,048,104,018 $F_p$ multiplications, 41,702,654 group operations and 369,021 modular row operations. This is an exact group-oracle pilot, not an $S_5$ or calibrated rho result.

## 1.9. Dense rows from one residual

The previous pilot retained one relation per residual. The new registered extractor kept every rank-raising decomposition of the **first** residual and fixed the public generator's factor log to 1. For a fixed residual, all unanchored row differences annihilate the nonzero factor-log vector and have target coordinate zero, so their rank is at most $m$ for $m+1$ unknowns. That ceiling was attained: 6/7 at p=13 and 52/53 at p=37. After removing the known column, the first residual had full rank on both curves. The independent point-keyed oracle verified all first-query decompositions, selected terms, rank and answer; A/A non-timing diffs are empty.

| $p$ | queries old → dense | accepted / candidates scanned | old → dense counted $F_pM$ | old → dense group ops |
|---|---|---|---|---|
| 13 | 7 → 1 | 6 / 6 | 8,096,696 → 2,045,670 | 81,076 → 18,609 |
| 37 | 53 → 1 | 52 / 115 | 4,048,104,018 → 119,740,945 | 41,702,654 → 1,204,356 |



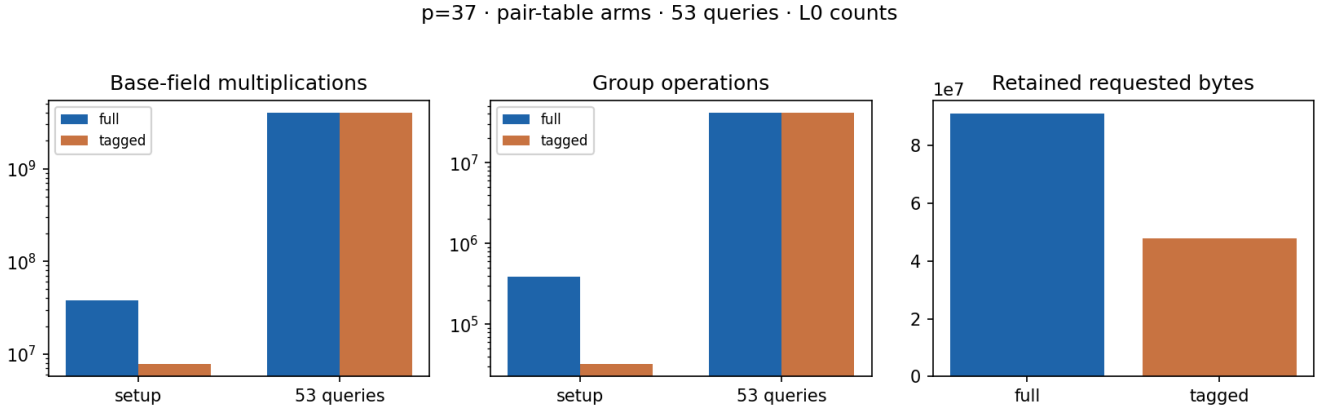

The p=37 counted one-target field and group paths fall 33.807-fold and 34.627-fold. This combines dense row retention with a public known-log anchor on the twist-folded 2D base; the reduction is not attributable to twist symmetry alone. The exact group oracle remains far from matched 24-fold rho and does not price polynomial relation finding, hash CPU, allocator work, isolated native time or a GAE band. The derivation, certified signed-count polynomial (`IDC1hf0237ddd316cd9d8`), raw records, independent audits, and limitations are in `QUARTIC_TWIST24_REPORT.md`.

For $B$ signed physical factors, a uniform residual has $16\frac{\binom{B}{4}}{r} \approx \left(\frac{2}{3}\right)\frac{B^4}{r}$ expected decompositions. With $m = \frac{B}{12}$ folded columns, the expected count reaches $m$ near $B^3 \approx \frac{r}{8}$; this is an occupancy heuristic, not a rank guarantee. The explicit pair oracle then builds $\Theta(B^2) = \Theta\left(r^{\frac{2}{3}}\right)$ entries with one group addition each, exceeding constant-automorphism rho's $\Theta\left(r^{\frac{1}{2}}\right)$ exponent. A 12-fold orbit quotient changes only the constant. Polynomial and implicit-table relation finders are outside this projection.

### 1.10. Full-stream tagged-table boundary

On the **old** 53-residual p=37 full-2D trace, direct-posting and full signed-pair tables returned identical complete sets for all 2,833,126 decompositions. Every tagged posting was checked by a separate group addition; A/A non-timing records matched. Query arithmetic was identical: 4,003,562,862 $F_p$ multiplications and 41,273,856 group operations in each arm.

| arm | setup $F_pM$ | setup group ops | retained requested bytes |
|---|---|---|---|
| full | 37,708,944 | 388,752 | 91,007,248 |
| tagged direct | 7,796,605 | 32,396 | 47,721,464 |

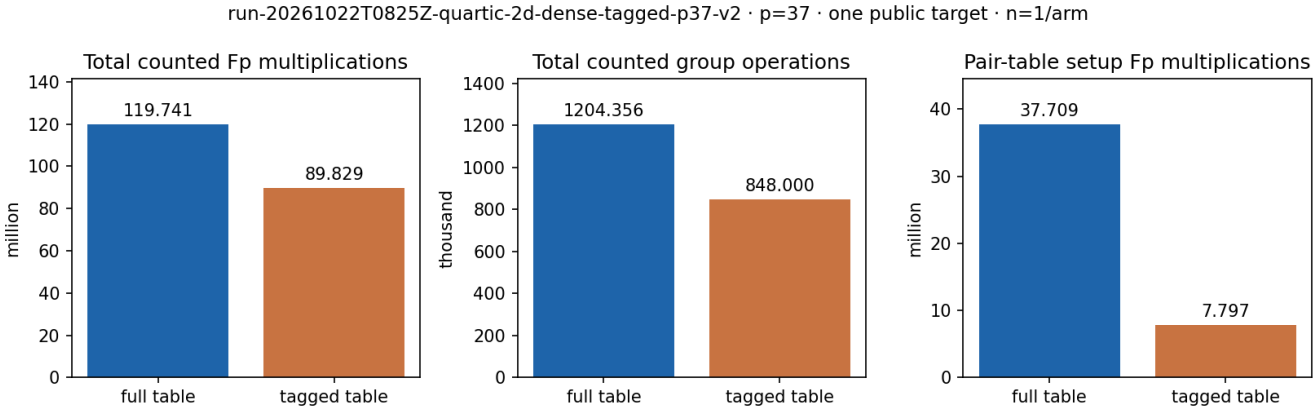



Setup plus 53 queries cost 4,041,271,806 versus 4,011,359,467 counted $F_p$ multiplications, a 1.00746-fold reduction. The 4.84-fold table-setup gain is diluted by the query stream. Its L0 wall time is not isolated; the tagged arm was slower in that shared-host observation. Dense extraction reduces the number of quadratic-size queries and gives the larger counted gain. Neither result establishes a polynomial or rho speedup.

## 1.11. Combined tagged index and dense rows in one producer

The registered p=37 combined producer charged model/base/input/eigen/map, a direct-posting tagged pair index, the first seed-zero residual, anchored rank selection, matrix solve and all answer checks in one exact group-oracle path. It used the same 624-point base and 52 folded columns as the earlier full-table dense arm. The tagged and separately charged full reference returned identical complete sets of 53,551 signed four-distinct-factor decompositions. The first 115 candidates supplied anchored rank 52/52 and the target scalar 271. The independent point-keyed auditor enumerated all 1,873,633 subgroup logs, replayed the complete hit count and selected rows, and independently recovered 271; the final-source A/A and original/final-source non-timing diffs are empty.

| one-target counted path | full-table dense | tagged dense | full/tagged |
|---|---|---|---|
| $F_p$ multiplications | 119,740,945 | 89,828,606 | 1.333 |
| group operations | 1,204,356 | 848,000 | 1.420 |
| table setup $F_pM$ | 37,708,944 | 7,796,605 | 4.837 |
| table setup group ops | 388,752 | 32,396 | 12.000 |



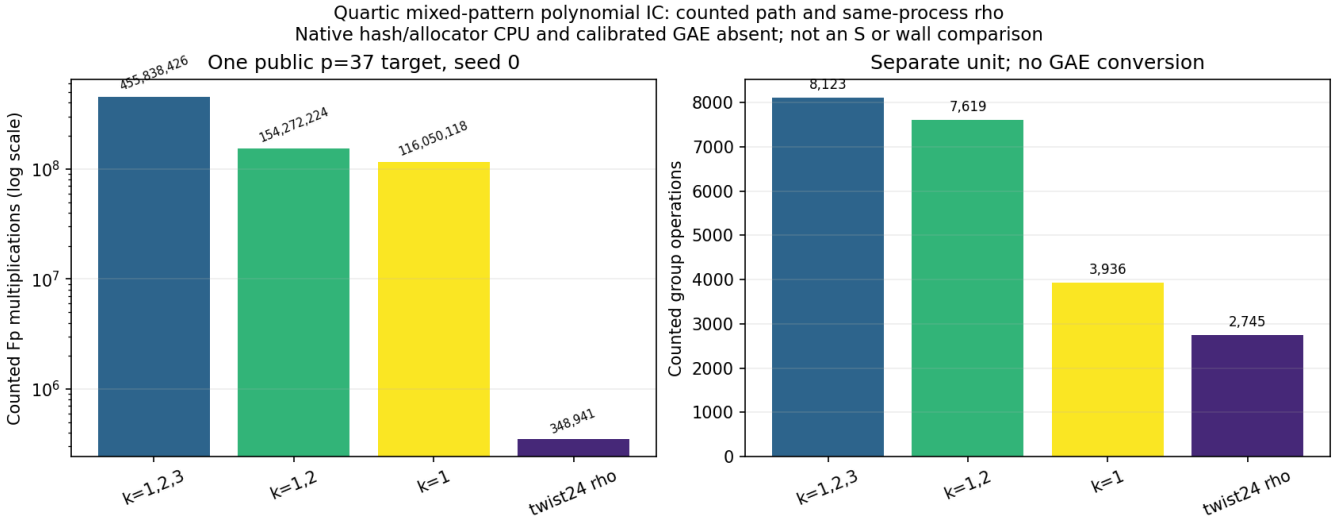

The total deltas of 29,912,339 $F_pM$ and 356,356 group operations equal the table-setup deltas exactly; all other counted phases match. The tagged table retained 47,721,464 requested bytes and the producer peaked 78,834,528 requested bytes above pre-table allocation. Its first-residual stream recorded 777,503 hash lookups and 324,839 candidate postings. The anchored matrix needed 32,453 modular operations; 17,251,447 additional operations checked an unanchored-rank diagnostic, so the shared-host L0 wall observation of 639.113 ms is not an isolated anchored-path time. Relative to the older 53-query full-table pilot, the combined counted ratios are 45.065-fold $F_pM$ and 49.178-fold group operations. This larger difference also includes generic dense-row extraction and a public known-log anchor, not only twist symmetry.

The exact-oracle path still costs about 257 times the earlier same-target 24-fold rho $F_pM$ count and 309 times its group-operation count, as separate cross-run diagnostics. Polynomial $S_5$ decomposition enumeration, hash/allocator CPU, isolated timing and a matched in-process 0/0.1/1 GAE band remain unmeasured. The full source-linked report, raw records, run/A-A PDFs, audit and protocol are in the study directory.

## 1.12. Mixed-component polynomial relation finding

The later registered p=37 tagged-line experiment uses the same 36 line-0 and 12 line-2 physical factors. The twist and cubic action fold them into four factor-log columns. One signed line-2 factor can be peeled, leaving an $S_4$ polynomial in three symmetric line-0 abscissae. On the frozen seed-zero public target $[271]G$, four distinct k=1 signed relations in eight residuals reach anchored rank four and recover the target. An independent scalar-keyed auditor enumerated all 1,873,633 subgroup logs, replayed every processed signed set and checked the factor logs, target and rho.

| polynomial patterns | residuals | counted IC $F_pM$ | IC group operations |
|---|---|---|---|
| k=1,2,3 | 7 | 455,838,426 | 8,123 |
| k=1,2 | 7 | 154,272,224 | 7,619 |
| k=1 | 8 | 116,050,118 | 3,936 |

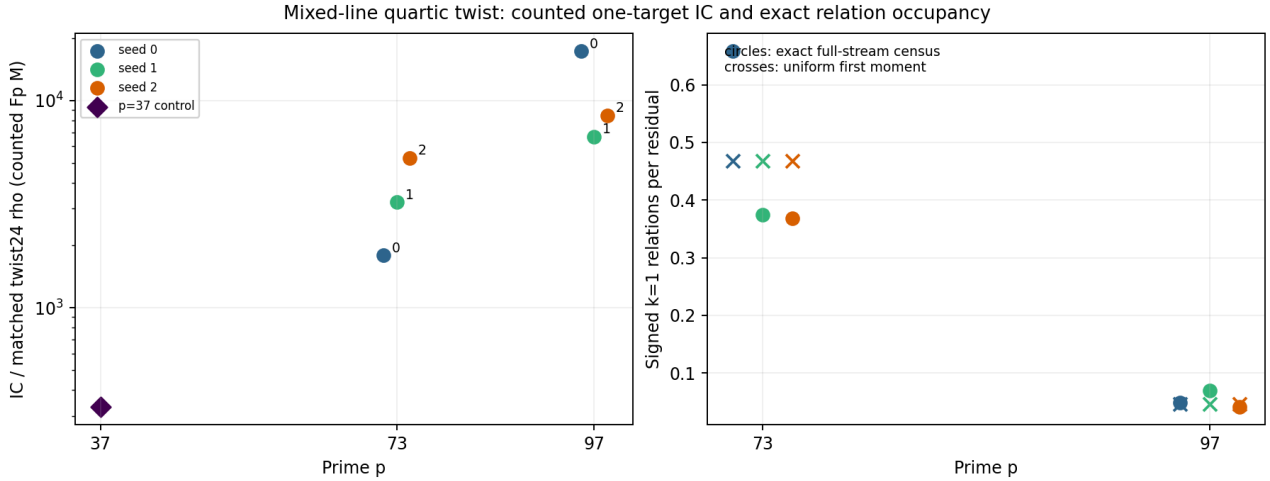



The matched same-process twist24 rho used 348,941 counted $F_pM$ and 2,745 group operations. In their separate units, the restricted k=1 path uses 332.6 and 1.434 times those counts. No calibrated $S$ or isolated wall-speed ratio is available: hash/tree and allocator CPU, requested peak bytes, rho hash/table CPU, external same-input Gröbner comparison and ABAB isolation are missing. The all-one-line k=0 S5 arm missed one verified relation; k=4 had a failed system, and degree-30/31 retries failed to repair the native Macaulay border. This result is a complete **restricted-pattern** polynomial IC solve on one toy instance, not a complete union S5 or rho crossover. The full algebra, raw evidence, A/A controls, failure records and graphs are in `MIXED_POLYNOMIAL_REPORT.md` and its own PDF.

## 1.13. Larger-curve mixed-pattern scaling and complete-stream occupancy

The next preregistered panel uses the identical mixed-line physical base and frozen residual streams at $p = 73$ and $p = 97$, with three independent one-target seeds each. The 24-action folds their 96 and 72 physical factors to eight and six log columns respectively. All six polynomial paths reach anchored full rank and recover $[271]G$. Five processed signed-relation sets exactly match an independent point-keyed group oracle. One $p = 73$ seed-one $S_4$ system fails and misses two verified relations, although the target is still recovered. That run is explicitly partial.

| $p$/seed | IC residuals | exact/poly signed sets | anchored rank | IC/rho counted $F_pM$ |
|---|---|---|---|---|
| 73/0 | 15 | 8/8 | 8/8 | 1,788 |
| 73/1 | 20 | 11/9 | 8/8 | 3,225 |
| 73/2 | 43 | 15/15 | 8/8 | 5,283 |
| 97/0 | 310 | 16/16 | 6/6 | 17,249 |
| 97/1 | 75 | 6/6 | 6/6 | 6,695 |
| 97/2 | 150 | 6/6 | 6/6 | 8,461 |




The separately registered **entire-stream** census counted 167 signed k=1 relations in 359 p=73 residuals and 79 in 1,497 p=97 residuals. In the p=97 streams, 1,423 residuals have no k=1 relation. The point-keyed group oracle and independent BSGS-derived scalar-keyed oracle agree on all 1,856 per-query signed-set digests, including after the polynomial arm stops. Its mutation control rejects one changed count at query zero. This identifies low physical k=1 occupancy as the source of the long p=97 collection paths, rather than a hidden polynomial miss on processed p=97 queries.

For each of the 42 selected polynomial relations, all 24 twist/cubic transforms of its folded matrix row are scalar multiples of that row: 1,008 exact row and right-hand-side checks and 12,096 factor-image closure checks pass. Thus the 24 action reduces unknown columns and supports component tagging but does not create 24 independent equations. The counted IC/rho field ratios above omit hash/tree and allocator CPU, peak bytes, calibrated group-arithmetic bands, isolated timing and same-input msolve/Magma. No priced whole-method speed or asymptotic conclusion follows. The source-linked full report and all run-level PDFs are in `K1_SCALING_REPORT.md`, `K1_SCALE_RUNS.md`, and `K1_CENSUS_RUNS.md`.